

Shuvomoy Das Gupta

CONTACT	 6100 Main St MS-134, Houston, TX 77005, USA  https://shuvomoy.github.io/  sd158@rice.edu
CITIZENSHIP	Canada
RESEARCH INTERESTS	Optimization, Game Theory, Computation, Transportation
CURRENT POSITION	Rice University, Houston, TX, USA 2025–Present <i>Assistant Professor, Computational Applied Mathematics & Operations Research</i>
ACADEMIC EXPERIENCE	Columbia University, New York, NY, USA 2024–2025 <i>Postdoctoral Research Scientist, Industrial Engineering & Operations Research</i> HOSTS: Garud Iyengar & Christian Kroer Worked on designing optimal algorithms for large-scale game solving.
INDUSTRY EXPERIENCE	Thales Canada Inc., Toronto, Canada 2016–2018 <i>Researcher, Research & Technology Department</i> Worked on real-time embedded optimization and sensor fusion algorithms in autonomous transportation systems.
EDUCATION	Massachusetts Institute of Technology, Cambridge, USA 2019 – 2024 Ph.D. in Operations Research THESIS: Advances in Computer-Assisted Design and Analysis of First-Order Optimization Methods and Related Problems PDF: https://dspace.mit.edu/handle/1721.1/155495 ADVISORS: Robert M. Freund & Bart P.G. Van Parys University of Toronto, Toronto, Canada 2016 Master of Applied Science in Electrical and Computer Engineering THESIS: Optimization Models for Energy-Efficient Railway Timetables PDF: https://utoronto.scholaris.ca/items/b89b9f6b-f9db-4757-8971-6d1563f779fd ADVISOR: Lacra Pavel
GRANTS, AWARDS AND HONORS	AFOSR Grant: “Computer-Assisted Design of Provably Fastest Algorithms”. Co-PI. \$600,000. (my share: \$300,000) 2025–2028 Winner, INFORMS Computing Society Student Paper Award 2024 Honorable Mention & Finalist, INFORMS Nicholson Student Paper Competition 2024 Honorable Mention, MIT Operations Research Center Best Student Paper Award 2024

**PUBLISHED
JOURNAL
PAPERS**

Uijeong Jang, Shuvomoy Das Gupta, and Ernest K. Ryu, “**Computer-Assisted Design of Accelerated Composite Optimization Methods: OptISTA**,” published in *Mathematical Programming Series A* 219 (2026): 391–499.
PDF: <https://arxiv.org/pdf/2305.15704.pdf>

Shuvomoy Das Gupta, Bart P.G. Van Parys, and Ernest K. Ryu, “**Branch-and-Bound Performance Estimation Programming: A Unified Methodology for Constructing Optimal Optimization Methods**,” published in *Mathematical Programming Series A*, 2024.
PDF: <https://arxiv.org/pdf/2203.07305.pdf>

Shuvomoy Das Gupta, Robert M. Freund, Xu Andy Sun, Adrien B. Taylor, “**Nonlinear Conjugate Gradient Methods: Worst-case Convergence Rates via Computer-assisted Analyses**,” published in *Mathematical Programming Series A*, 2024.
PDF: <https://arxiv.org/pdf/2301.01530.pdf>

Shuvomoy Das Gupta, Bartolomeo Stellato, and Bart P.G. Van Parys, “**Exterior-point Optimization for Sparse and Low-rank Optimization**,” published in the *Journal of Optimization Theory and Applications*, 2024.
PDF: <https://arxiv.org/pdf/2011.04552.pdf>

Shuvomoy Das Gupta and Lacra Pavel, “**On Seeking Efficient Pareto Optimal Points in Multi-Player Minimum Cost Flow Problems with Application to Transportation Systems**,” published in the *Journal of Global Optimization* 74 (2019): 523–548.
PDF: <https://arxiv.org/pdf/1805.11750.pdf>

Shuvomoy Das Gupta, J. Kevin Tobin, and Lacra Pavel, “**A Two-Step Linear Programming Model for Energy-Efficient Timetables in Metro Railway Networks**,” published in *Transportation Research Part B: Methodological* 93 (2016): 57–74.
PDF: <https://arxiv.org/pdf/1506.08243.pdf>

**PUBLISHED
CONFERENCE
PAPERS**

Tianlong Nan, Shuvomoy Das Gupta, Garud Iyengar, Christian Kroer, “**On the $O(1/T)$ Convergence of Alternating Gradient Descent–Ascent in Bilinear Games**,” published in *ICLR*, 2026.
PDF: <https://arxiv.org/pdf/2510.03855>

Jakub Cerny, Shuvomoy Das Gupta, and Christian Kroer, “**Spatial Branch-and-Bound for Computing Multiplayer Nash Equilibrium**,” published in *AAAI*, 2026.
PDF: <https://arxiv.org/pdf/2508.10204>

Shuvomoy Das Gupta, “**On Convergence of Heuristics Based on Douglas-Rachford Splitting and ADMM to Minimize Convex Functions over Nonconvex Sets**,” published in the *proceedings of the 56th Allerton Conference on Communication, Control, and Computing*, University of Illinois at Urbana-Champaign, IL, USA, October 2018.
PDF: https://shuvomoy.github.io/Papers/Allerton_2018.pdf

Shuvomoy Das Gupta and Laca Pavel, “**Multi-player Minimum Cost Flow Problems with Nonconvex Costs and Integer Flows**,” published in the *proceedings of the 55th IEEE Conference on Decision and Control*, Las Vegas, NV, USA, December 12–14, 2016.

PDF: https://shuvomoy.github.io/Papers/Multi-player_minimum_cost_flow_problems_with_nonconvex_costs_and_integer_flows.pdf

Shuvomoy Das Gupta, Laca Pavel, and J. Kevin Tobin, “**An Optimization Model to Utilize Regenerative Braking Energy in a Railway Network**,” published in the *proceedings of 2015 American Control Conference (ACC)*, Chicago, IL, USA, July 2015.

PDF: https://shuvomoy.github.io/Papers/An_Optimization_Model_to_Utilize_Regenerative_Braking_Energy_in_a_Railway_Network.pdf

PREPRINTS

Tianlong Nan, Garud Iyengar, Christian Kroer, and Shuvomoy Das Gupta, “**AltGDA Achieves Global $O(1/T)$ Ergodic Convergence in Matrix Games**,” arXiv preprint, 2026.

PDF: <https://arxiv.org/pdf/2609.32995>

CODE: <https://github.com/Shuvomoy/AltGDA-code>

Heechang Kim, Ernest K. Ryu, and Shuvomoy Das Gupta, “**A Domain-Specific Harness for End-to-End Automation of Optimization Research**,” arXiv preprint, 2026.

PDF: <https://arxiv.org/pdf/2608.07407>

CODE: <https://github.com/Shuvomoy/AutoOPT>

Manu Upadhyaya, Shuvomoy Das Gupta, Adrien B. Taylor, Sebastian Banert, and Pontus Giselsson, “**The AutoLyap Software Suite for Computer-Assisted Lyapunov Analyses of First-Order Methods**,” arXiv preprint, 2026.

PDF: <https://arxiv.org/pdf/2506.24076>

TEACHING

CMOR 537: Computer-assisted Algorithm Design for Optimization and Machine Learning, Rice Spring 2026

Instructor. New graduate course for its inaugural offering.

CMOR 467/567: Optimization for Energy Systems, Rice Fall 2025, 2026

Instructor. I have designed and launched this new course for advanced undergraduate and graduate students.

6.7220: Nonlinear Optimization, MIT Spring 2023

Teaching Assistant. This is MIT’s main doctoral course in optimization.

RATING: 6.9/7.0

15.S60: Computing in Optimization and Statistics, MIT Winter 2023

Instructor. I taught the ORC’s required three-hour module on advanced methods in computational optimization.

RATING: 7/7

15.S60: Computing in Optimization and Statistics, MIT Winter 2022

Instructor. I taught the ORC’s required three-hour module on advanced methods in computational optimization.

RATING: 6.8/7

15.S08: Optimization of Energy Systems, MIT Spring 2022
Teaching Assistant. This is a graduate course in power systems modeling and optimization.
 RATING: 6.0/7.0

ADVISING AND MENTORING

Ph.D. students

- Qiuyu Fan (Rice University) 2025–

Undergraduate students

- Heechang Kim (Seoul National University) 2025–
- Uijeong Jang (Seoul National University; current: Ph.D. student at UCLA) 2022–23

TALKS

A Domain-Specific Harness for End-to-End Automation of Optimization Research
 MOPTA 2026, Lehigh University, Bethlehem, PA 2026

PEPit.jl: Computer-Assisted Worst-Case Analysis of First-Order Optimization Algorithms in Julia
 JuMP-dev 2026, Edinburgh, Scotland 2026

On the $O(1/T)$ Convergence of Alternating Gradient Descent–Ascent in Bilinear Games
 SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom 2026
 Mixed Integer Programming Workshop 2026, University of Connecticut – Stamford Campus 2026
 INFORMS Annual Meeting, Atlanta, GA 2025

Computer-Assisted Design of Provably Fastest Algorithms
 Invited talk, New Jersey Institute of Technology, New York, NY 2025

Nonlinear Conjugate Gradient Methods: Worst-case Convergence Rates via Computer-assisted Analyses
 ICCOPT, Los Angeles, CA 2025
 INFORMS Annual Meeting, Seattle, WA 2024

BnB-PEP: A Unified Methodology for Constructing Optimal Optimization Methods
 INFORMS Annual Meeting, Phoenix, AZ 2023
 SIAM Conference on Optimization (OP23), Seattle, Washington 2023
 UTOrg Seminar, University of Toronto, Toronto, Canada 2023
 International Conference on Continuous Optimization, Bethlehem, PA 2022
 MIT Data Science Lab Seminar 2022

Design and Analysis of First-Order Methods via Nonconvex QCQP Frameworks
 One of just four invited “long talks” at the 1st Workshop on Performance Estimation, UCLouvain, Belgium 2023

Energy-Optimal Timetable Design for Sustainable Metro Railway Networks
 INFORMS Annual Meeting, Phoenix, AZ 2023
 33rd Annual POMS Conference, Orlando, FL 2023
 2023 MIT Energy Initiative Annual Research Conference 2023

NExOS.jl for Nonconvex Exterior-point Optimization
 JuliaCon 2021

Exterior-point Optimization for Sparse and Low-rank Optimization
 Seoul National University 2020
 INFORMS Annual Meeting (virtual) 2020
 MIT LIDS student conference 2021

On Convergence of Heuristics Based on Douglas-Rachford Splitting and ADMM to Minimize Convex Functions over Nonconvex Sets
 56th Allerton Conference on Communication, Control, and Computing, Monticello, IL 2018

Multi-Player Minimum Cost Flow Problems with Nonconvex Costs and Integer Flows
 55th IEEE Conference on Decision and Control, Las Vegas, NV 2016

On seeking efficient Pareto optimal points in multi-player minimum cost flow problems
 MIT LIDS student conference 2020

SERVICE

Chair, Rice CMOR Colloquium Organization Committee 2026–2027
 Coordinating the departmental colloquium program, including speaker invitations, scheduling, hosts, and seminar logistics.

Member, INFORMS George Nicholson Student Paper Competition 2026–2027

Member, Rice CMOR Undergraduate Committee 2026–2027
 Supporting undergraduate curriculum, advising, major/minor requirements, and student-facing program resources.

Member, Rice CMOR Graduate Committee 2025–2026
 Supported graduate program administration, including student advising, degree requirements, qualifying exams, and academic progress.

Organizer, Friends of Optimization Seminar Series 2024–
 LINK: <https://sites.google.com/view/friends-of-optimization>

Reviewer for *Mathematical Programming, Management Science, Transportation Research Part B: Methodological, IEEE Transactions on Control of Network Systems, American Control Conference, IEEE Transactions on Intelligent Transportation Systems, IEEE Transactions on Automatic Control*

Session Chair, INFORMS Annual Meeting 2022–2025

SOFTWARE

[1] **AltGDA-code**
 Julia experiments and Lean 4 formalization for the global ergodic convergence of AltGDA in matrix games.
 LINK: <https://github.com/Shuvomoy/AltGDA-code>

[2] **AutoOPT**

Domain-specific harness for end-to-end automation of optimization research through numerical algorithm design, symbolic discovery, and Lean verification.

LINK: <https://github.com/Shuvomoy/AutoOPT>

[3] **PEPit.jl**

Native Julia implementation of the Performance Estimation Programming (PEP) methodology.

LINK: <https://github.com/PerformanceEstimation/PEPit.jl>

[4] **AutoLyap.jl**

Native Julia implementation of the AutoLyap methodology.

LINK: <https://github.com/AutoLyap/AutoLyap.jl>

[5] **BnB-PEP**

Computes optimal first-order algorithms for different convex and nonconvex setups

LINK: <https://github.com/Shuvomoy/BnB-PEP-code>

[6] **NCG-PEP**

Computes worst-case convergence rates of nonlinear conjugate gradient methods

LINK: <https://github.com/Shuvomoy/NCG-PEP-code>

[7] **NExOS**

Implements the Nonconvex Exterior-point Optimization Solver (NExOS) algorithm for solving low-rank and sparse optimization problems

LINK: <https://github.com/Shuvomoy/NExOS.jl>

LANGUAGES

Fluent in

English, Bengali, Hindi, Urdu

Proficient in

Julia, C, C++, MATLAB, Mathematica

OTHER

I enjoy playing cricket, reading novels, cooking, and blogging at <https://shuvomoy.github.io/blogs/>.

**MEDIA
COVERAGE**

“Risky Giant Steps Can Solve Optimization Problems Faster” August, 2023 by Allison Parshall in *Quanta Magazine*

I was interviewed and quoted in the article along with my paper being cited as the main inspiration for the discovery of long step gradient descent by Ben Grimmer. Also publicized in the *Nautilus Quarterly Magazine* and in the Chinese magazine *Heart of the Machine*.

URL: <https://www.quantamagazine.org/risky-giant-steps-can-solve-optimization-problems-faster-20230811/>